**Interview Questions**

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**1. What is APT?**

**APT** (Advanced Package Tool) is the default command-line package management system used in Debian-based Linux distributions (like Ubuntu, Debian, and Linux Mint). It simplifies managing software by handling installation, updates, uninstallation, and automatically resolving package dependencies.

**2. Difference between apt update and apt upgrade?**

- **apt update**: Refreshes your local package index/cache by fetching the latest repository metadata from the internet. It *does not* update any installed software; it only checks what new versions are available.
- **apt upgrade**: Downloads and installs newer versions of all currently installed packages on your system, based on the lists fetched by apt update.

**3. What does df -h display?**

df stands for **disk free**. It displays information about available and used disk space across all mounted filesystems. The **-h** flag stands for "human-readable," converting raw byte counts into intuitive units like KB, MB, or GB (e.g., 45G instead of 47185920).

**4. Difference between df -h and lsblk?**

- **df -h**: Focuses on **filesystems**—it shows how much disk space is used, available, and mounted *inside* active filesystems.
- **lsblk**: Focuses on **block devices** (hard drives, SSDs, partitions, USBs) and displays them in a tree structure, showing device types, sizes, and relationships regardless of whether they are currently full or empty.

**5. What is a UUID?**

A **UUID** (Universally Unique Identifier) is a unique 128-bit alphanumeric string assigned to a partition or filesystem when it is created. Linux systems use UUIDs (instead of device names like /dev/sdb1) to

reliably identify and mount drives, ensuring that storage configurations do not break if hardware connections or drive letters change.

**6. What is mounting in Linux?**

Mounting is the process by which the operating system makes files and directories on a storage device (such as a hard drive, partition, USB stick, or network share) accessible to users by attaching it to a specific directory (called a **mount point**) within the main file system tree.

**7. Difference between mount and umount?**

- **mount**: Attaches a filesystem or storage device to a designated directory path so it can be accessed.
- **umount**: Safely detaches a filesystem from its directory tree, flushing any cached data to disk so the device can be safely removed or uninstalled without data corruption.

**8. What is /etc/fstab?**

/etc/fstab (File System Table) is a system configuration file used to define how disk partitions, various filesystems, and storage devices should be mounted automatically by the operating system during the boot process.

**9. Difference between .tar and .tar.gz?**

- **.tar**: A tape archive file that bundles multiple files and directories into a single file for easy storage or transport, but **does not apply compression**.
- **.tar.gz**: A .tar archive that has been additionally compressed using **gzip**, significantly reducing the overall file size.

**10. Why is rsync preferred in production?**

rsync is widely preferred in production environments for backups and file synchronization due to several key advantages:

- **Delta-transfer algorithm**: It only transfers the changes or modified blocks within files rather than copying entire files over again, saving bandwidth and time.
- **Preservation of attributes**: It can preserve file permissions, ownership, timestamps, and symbolic links (using the -a or archive flag).
- **Network efficiency & security**: It can run securely over SSH and handles interrupted transfers smoothly.